Jorge Luis Castro García

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Professional Summary

Senior Software Engineer with 9 years of experience delivering high-impact, scalable solutions for the semiconductor and high-tech industries. Proven track record in designing and maintaining complex systems that integrate hardware and software, accomplished through the creation of detailed feature specifications, workflow automation, and strategic coordination with R&D, manufacturing, and business stakeholders to gather requirements and deliver results. Recognized for high-level technical leadership and owning end-to-end feature development in ambiguous environments. Adept at bridging technical complexity with business while continuously identifying opportunities to optimize processes and modernize legacy systems.

Experience

October 2023 – Present: Full Stack Software Engineer (National Instruments – Emerson T&M)

- Leading initiatives to align STS firmware update processes with EU Cyber Resilience Act (CRA) requirements, ensuring regulatory compliance and strengthening system security for European market readiness.
- Designed and implemented an AI-powered diagnostic solution using .NET 9.0, LLMs, and Retrieval-Augmented Generation (RAG) to analyze system logs, drastically reducing manual troubleshooting time.
- Designed and maintained backend services and frontend applications using C#, MySQL, JavaScript, Python, and LabVIEW, supporting monitoring and maintenance platforms for semiconductor test systems used in high-reliability manufacturing environments.
- Contributed key features to the Semiconductor Test System (STS) Maintenance Software, a C#-based platform for monitoring and maintaining system health, including integration of in-development hardware devices to expand diagnostic and calibration capabilities.
- Achieved an 80% reduction in manual hardware data entry by developing a Python-based automation tool for R&D libraries, significantly improving data reliability.
- Acted as Technical Lead for two business-critical internal tools, driving architecture decisions, code reviews, and feature delivery while coordinating with cross-functional teams to ensure scalability, reliability, and alignment with product requirements.
- Collaborated with cross-functional global teams (Sales, Manufacturing, and R&D) to drive continuous improvement in an iterative development environment, ensuring alignment across product lifecycle stages.

November 2019 – October 2023: Technical Support Engineer (National Instruments – Emerson T&M)

- Diagnosed and resolved complex hardware-software integration issues, performing root-cause analysis and collaborating with R&D, Sales, and Product Planning to deliver short and long-term fixes.
- Provided technical support for NI data acquisition systems (PXI, cDAQ, and Linux Real-Time targets) using diverse programming languages, including C#, Python, C/C++, and LabVIEW.
- Acted as Subject Matter Expert for key software platforms: LabVIEW, LabWindows/CVI (ANSI C), Measurement Studio (C#/VB.NET), driving knowledge-sharing and mentoring.
- Developed LabVIEW Connectivity Learning Path, a comprehensive course on building robust network communication and data integration solutions with LabVIEW.
- Trained support engineers in C# and ANSI C, expanding team proficiency in text-based programming and improving customer support for complex measurement and automation projects.
- Maintained a 4.8/5.0 Customer Satisfaction (CSAT) average, earning recognition for high-quality resolution of complex system-level escalations. Exceeded productivity benchmarks with an efficiency of 3.14 hr/case, 28% faster than the department average.

January 2017 – November 2019: Test Development Engineer (Teradyne)

- Designed and executed diagnostic, regression, and performance tests for semiconductor platforms (C#, C++, VB.NET).
- Improved engineering efficiency by building a Visual Studio extension that automated test project creation through customized templates.
- Automated semiconductor test workflows by designing and building a C#-based results management application with integrated server uploads and real-time user notifications.
- Resolved internal and external customer issues with core software architecture and user interface, ensuring timely solutions by collaborating with local and US-based teams using Agile methodologies.

March – June 2016: Intern (Continental AG, Germany)

- Conducted geometric verification of valve blocks in vehicle braking systems using coordinate measuring machines; analyzed and reported deviations from nominal specifications.
- Tested quality control instruments and ensured they met tolerance parameters through detailed reporting.

Education

Master of Science in Computer Science (2026 – Present)
Georgia Institute of Technology. OMSCS Program. Specialization in Artificial Intelligence.

Mechatronics Engineering Licentiate Degree (2011 – 2016)
Costa Rica Institute of Technology. Graduated with honors. GPA: 90.51/100

Electronics & Technology Management. Exchange program supported by the Ernst-Mach scholarship (2014)
Fachhochschule Joanneum, Austria

Languages

- Spanish: Native language.
- English: Advanced level.
- German: Intermediate level.

Technical Skills

Backend: C/C++, C#, VB.NET, LabVIEW, Python, JavaScript, Matlab, Node.js, SQL.
Frontend: HTML/CSS, React, Blazor.
Databases: MySQL Server, Qdrant.
Software tools: Git, SourceTree, IBM ClearCase, Azure DevOps, Visual Studio, VS Code, Matlab, MySQL, LabVIEW, LabWindows/CVI, Measurement Studio, TestStand, FlexLogger, SystemLink, DIAdem, Docker, xUnit.